

Global Facilities - Design & Construction Standard - General - OSM (Off-Site Manufacturing/Modularization)

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1.0 PURPOSE

The purpose of this OSM (Off-Site Manufacturing/Modularization) Standard is to provide the framework for execution of the OSM portion of the project. We refer to OSM to represent all off site, modular, and prefabrication/pre-assembly. The document provides guidelines for design coordination, procurement, manufacturing, quality control, logistics and construction install coordination, and delivery of infrastructure that will be manufactured offsite.

This Standard provides an overview of all the common elements that will be outlined by the individual vendors in their respective Execution Plans. This is a generic Standard that covers all OSM equipment scope on the project. All Micron standards and specifications still apply to OSM scope.

This document also provides coordination guidelines between Micron, Micron's Modular Consultant, and the Construction Management Team, OSM vendor, and Micron's Design A/E. The OSM systems are considered as those that are procured as complete systems directly from manufacturers. This includes large stand-alone modular building infrastructure as well as other smaller OSM items such as modular pump skids, etc. that will be procured directly from manufacturers/contractors as complete systems or potentially self-performed OSM by the local trades.

This standard is intended to clarify the roles and responsibilities of the parties involved in the project. This document should not be interpreted as a modification or amendment to any of the contractual relationships. The procedures in this Standard may be modified and updated to suit the requirements of the specific OSM project.

2.0 SCOPE

The OSM scope development process will begin with the identification of infrastructure that can be fabricated offsite.

The opportunities will be vetted by the technical working groups and will be accounted for at various stages of the overall design process.

A separate Modular/OSM Design Work Group shall be established with representatives from each of the major design discipline work groups with construction, procurement and schedule input that will determine the project execution for each opportunity. This workgroup will ensure that each of the OSM opportunities is supported with a design/procurement strategy that supports the overall construction schedule.

An OSM tracker (Figure 1 below) will track these opportunities on a regular basis through the design process. The tracker will be a working document that will collect and modify data related to new OSM opportunities in real-time. The working groups will develop all the relevant, required equipment specifications for the scope. Depending on the complexity of the OSM project, the workgroups may outline a demarcated scope on the design drawings that will be required to be procured and manufactured offsite. If that is the case, the OSM vendor will review the demarcated drawings and will execute the detail design (along with the fabrication scope). The Tracker is embedded in the appendix.

Figure 1 Module Tracker (Full document attached at the end of this standard)

Reference Number	OSM Opportunity Description	Design Package	IFC Date	Design Owner	Procurement Owner	CM Owner	Field Need Date	Main Discipline	System	FOAK (Y/N)	Expected Benefit	Execution Strategy	Supplier Pre-qual	RFP Date	PO Number	PO Date	Submittal Date	Delivery Date	Factory Test Owner	Shipping Owner	FAT Owner	Installation Owner	Commissioning Owner
1																							
2																							
3																							

3.0 Definitions and Acronyms

- **4D BIM** – A BIM model integrated with the construction schedule to create a visual representation of the build sequence. Synchro is a tool commonly used to create 4D models.
- **AIA**: American Institute of Architects
- **ANSI** - American National Standards Institute
- **A/E** - Architecture/Engineering
- **AHJ** - Authorities Having Jurisdictions
- **BB** – Base Build
- **BOD** - Basis of Design
- **BIM** - Building Information Modeling
- **CE** - Conformité Européenne
- **CFCI** – Contractor Furnish, Contractor Install
- **DfMA** - Design for Manufacturing and Assembly is design approach that focuses on ease of manufacture and efficiency of assembly. By simplifying the design of a product it is possible to manufacture and assemble it more efficiently, in the minimum time and at a lower cost.
 - Traditionally, DfMA has been applied to sectors such as the design of automotive and consumer products, both of which need to efficiently produce high quality products in large numbers. More recently, construction contractors have begun to adopt DfMA for the off-site prefabrication of construction components such as concrete floor slabs, structural columns, and beams, and so on.

- **DFM** - Design for Manufacturing involves designing for the ease of manufacture of a product's constituent parts. It is concerned with selecting the most cost-effective materials and processes to be used in production, and minimizing the complexity of the manufacturing operations.
- **DFA** - Design for Assembly involves design for a product's ease of assembly. It is concerned with reducing the product assembly cost and minimizing the number of assembly operations.
 - Both DFM and DFA seek to reduce material, overhead, and labor costs.
 - DFM, DFA, and DFMA are best applied with an integrated team including the designer and the builders.
- **DV** - Design Verification
- **EHS** - Environmental, Health, and Safety
- **Fabrication** - For the purposes of this standard, fabrication is defined as the offsite construction of modular scope (preassembly, prefabrication, OSM, volumetric, skids, etc.).
- **FOAK** - First of A Kind modules
- **FAT** - Functional Acceptance Test
- **GC** - General Contractor
- **GEO** - Geography
- **HDB** - Housing Development Board
- **Hz** - Hertz
- **IC - Industrialized Construction** is methodology that goes beyond prefabrication, offsite, and modular construction by leveraging industrial engineering principles to optimize the entire building process from design to completion. IC is often centered around a product driven design approach combined with a production schedule designed for flow.
- **IFC** - Issue for Construction
- **IFP** - Issue for Purchase
- **IMD** - Integrated Modular Design
- **I&C** - Instrumentation and Controls
- **Integrator** - For the purposes of this standard, the Integrator is defined as a fabrication contractor that assembles modules (volumetric, skid, OSM, etc.) incorporating equipment and materials from other suppliers. An example would be an electrical skid where the gear is purchased and assembled by the fabricating contractor (integrator). Other examples include hillier Plants, Compressor Buildings and E-houses.
- **ICC** - International Code Council
- **IMO** - International Maritime Organization
- **KPI** - Key Performance Indicators
- **LOD** - Level of Detail
- **LTA** - Land Transport Authority
- **Modularization** - Complete units of scope that require only installation and connection of input/output utilities at the site. Modular construction concepts generally require the involvement of the A/E. They can often change the build sequence.
- **MIMO** - Move-In/Move-out
- **MOM** - Ministry of Manpower
- **NEA** - National Environment Agency
- **OFCI** - Owner Furnish, Contractor Install
- **OSF** - Off-Site Fabrication is the manufacturing, planning, design, fabrication, and assembly of building elements at a location other than their final installed location to support the rapid speed of, and efficient construction of a permanent structure. Off-Site Fabrication generally requires coordination with

the engineer and/or architect of record. It can also substantially change the build and equipment purchase timing.

- **OSM** - Offsite Manufacturing
- **OAT** - Owners Acceptance Test
- **P-DFMA** - Platform Approach to Design for Manufacture and Assembly is the use of a set of digitally designed components across multiple types of built asset that are then used wherever possible, minimizing the need to design bespoke components for different types of asset.
- **Preassembly** – The joining of prefabricated components together away from the final building site. As a general assumption, preassembly does not require extensive A/E involvement or coordination.
- **Prefabrication** – An off-site manufacturing process that utilizes a specialized facility to build simple components away from the construction site. As a general assumption, prefabrication does not require extensive A/E involvement or coordination.
- **PPVC** - Prefabricated Prefinished Volumetric Construction (Singapore) is a construction method whereby free-standing 3-dimensional modules are completed with internal finishes, fixtures, and fittings in an off-site fabrication facility, before it is delivered and installed on-site.
- **P&ID** - Piping and Instrumentation Diagram
- **PFD** - Process Flow Diagram
- **PLC** - Programmable Logic Controller
- **PM** - Project Manager
- **PPVC** – Prefabricated Prefinished Volumetric Construction
- **QA** - Quality Assurance
- **QC** - Quality Control
- **RMF** - Raised Metal Flooring
- **RFP** - Request for Proposal
- **SCDF** – Singapore Civil Defense Force
- **SoR** - Schedule of Rates is a pre-priced list of typical tasks that can be used together to calculate the value of work to be completed
- **SOW** - Scope of Work
- **SITP** - Site Inspection and Test Plan
- **SOR** - Site Observation Report
- **Skid** – A modular assembly involving one or more equipment components. Skids are assembled on their own support structure and anchored into place.
- **Stick Built** – The traditional method of delivering construction where equipment, materials, and people are brought to the building site to complete the work in place.
- **T&C** - Terms and Conditions
- **URA** – Urban Redevelopment Authority
- **UL** - Underwriter Laboratories
- **Volumetric Construction** - modules constructed in a factory that are installed on site to form a cellular system (or used independently as a self-contained cell). Most volumetric modules are rooms (bathrooms, kitchens, jail cells, etc.). The delivery of services to volumetric modules are much different than traditional, stick built rooms.
 - Cellular systems are used for repetitive designs. Common uses include hotels, prisons, student halls and residential buildings. Self-contained cells are used mostly for specialized purposes where services are needed such as wet rooms, bathroom pods and service utility rooms. Once lifted into place, the modules are secured by several methods including bolted and doweled connections.
- **WG** - Work Group

- **FCT** - Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
- **GFTT** - Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
- **GFC** - Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
- **Global Facilities Central Team** - This is used interchangeably with Innovation & Integration Team
- **Global Facilities Construction Team** - This is used interchangeably with Global Construction Team
- **Global Facilities Operations Support Team** - This is used interchangeably with Global Facilities Operations Team
- **Global Facilities Planning Team** - This is used interchangeably with Global Facilities Line Planning Team

4.0 Roles & Responsibilities

4.1 For the Full A/E Design Approach the roles and responsibilities are generally:

Roles	Responsibilities
A/E	• Completes design per standard and contract documents • Coordinates BIM model with vendors and equipment suppliers until IFC • Reviews and approves submittals • Answers RFI's from builder and installer • Completes Design Verification per contract documents • Creates/maintains standard designs for modules
BB GC	• Leads procurement of modules (CFCI) • Coordinates delivery of modules (CFCI) • Coordinates FAT with vendors, Micron, and other suppliers as required (CFCI) • Coordinates delivery and installation of modules between module suppliers and local contractors • Provides staging as required for modules prior to installation • Coordinates BIM model post-IFC between module suppliers and local contractors • Provides module red line drawings to AE post installation • Responsible for commissioning of modules per contract documents • Responsible for site readiness prior to module delivery

Roles	Responsibilities
Modular Builder	- Constructs modules per contract documents - Submits shop drawings for Owner and A/E review prior to manufacturing - Coordinates delivery of modules with GC - Verifies equipment is compliant with submittals - Coordinates with AE for connections to buildings and facility systems - Submits module changes to AE/GC for approval - Submits final BIM model to GC for clash detection
Micron/Micron Modular Consultant	- Leads Modular WG - Pre-qualifies Modular suppliers - Leads procurement of modules (OFCI) - Coordinates delivery of modules (OFCI) - Coordinates FAT with vendors, Micron, and other suppliers as required (OFCI) - Leads detailed planning for module builds, shipping, and install with all stakeholders. - Owns modular standard and updates with new learnings - Transfers lessons to future projects

4.2 For modules designed by the modular builder/integrator the roles and responsibilities are generally:

Roles	Responsibilities
A/E	- Coordinates BIM model with vendors and equipment suppliers until IFC - Reviews and approves submittals - Answers RFI's from builder and installer - Completes Design Verification per contract documents - Creates/maintains standard designs for modules
BB GC	- Leads procurement of modules (CFCI) - Coordinates delivery of modules (CFCI) - Coordinates FAT with vendors, Micron, and other suppliers as required (CFCI) - Coordinates delivery and installation of modules between module suppliers and local contractors - Provides staging as required for modules prior to installation - Coordinates BIM model post-IFC between module suppliers and local contractors - Provides module red line drawings to AE post installation - Responsible for commissioning of modules per contract documents - Responsible for site readiness prior to module delivery

Roles	Responsibilities
Modular Builder	• Designs module per design standard and contract documents • Submits shop drawings for Owner and A/E review prior to manufacturing • Constructs modules per contract documents • Coordinates delivery of modules with GC • Verifies equipment is compliant with submittals • Coordinates with AE for connections to buildings and facility systems • Submits module changes to AE/GC for approval • Submits final BIM model to GC for clash detection
Micron/ Micron Modular Consultant	• Leads Modular WG • Completes design verification • Pre-qualifies Modular suppliers • Leads procurement of modules (OFCI) • Coordinates delivery of modules (OFCI) • Coordinates FAT with vendors, Micron, and other suppliers as required (OFCI) • Leads detailed planning for module builds, shipping, and install with all stakeholders. • Owns modular standard and updates with new learnings • Transfers lessons to future projects

Note: The project team will document the complete roles and responsibilities per module in the tracker (template attached to this document). It is extremely important to have clear definition on the roles for successful modular execution. "The success of a modular construction approach is contingent on how well information is shared between the various parties involved, especially the design team, the modular manufacturer, and the onsite construction management team." -AIA Modular Construction Guide (referenced below).

5.0 Safety

- 5.1 Modular construction has specific safety concerns to be addressed during design. Note, A/E is responsible for safety in design with regards to standard building codes and local Authorities Having Jurisdictions (AHJ) requirements. Contractors are responsible for construction safety means and methods.

Modular safety in design and construction should be considered but not be limited to:

- 5.1.1 **Codes:** Industry standard and local AHJ building codes. All design activities will be compliant with ICC/MBI 1200-2021 "Standard for Offsite Construction: Planning, Design, Fabrication and Assembly" as well as local codes and standards.
- 5.1.2 **Rigging** – Modules, skids, volumetrics, etc. are built offsite and moved into place. The center of gravity should be known and understood to prevent tipping. Rigging lugs should be designed to avoid improper rigging by an expert. Designers should seek to understand any rigging

requirements during the design of the modules. Designers should also avoid pinch points.

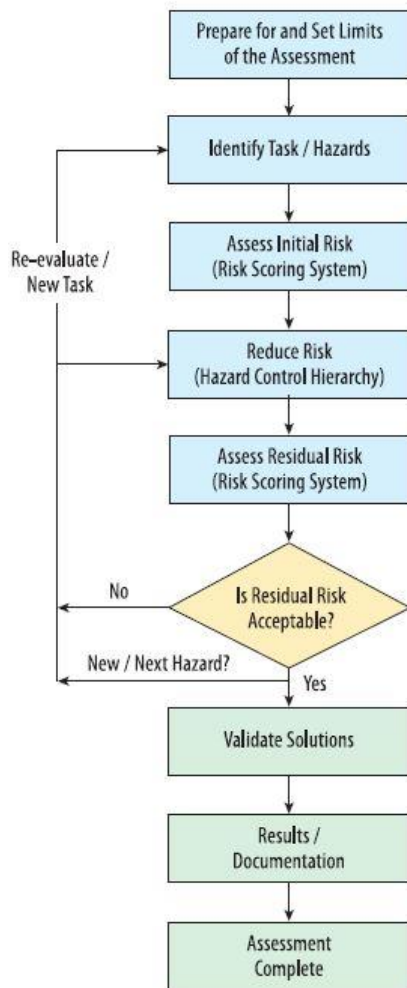
- 5.2 Safety Risk Assessments should be completed throughout the design process. During the risk assessments, manufacturing, installation, and operations hazards should be identified and mitigated using the Risk Assessment Process from ANSI B11.0 (Figure 2)
- 5.3 Modular construction provides benefits in both efficiency and safety. Logistically, smaller skids are easier to deliver and install. As a result, this creates fewer associated risks to rigging crews. The goal of modular construction is not to build the biggest skids possible, but to build a skid that meets the customer's needs, reduces scheduling impacts, and addresses the value of an incident-free installation.

Evaluate logistics for maximum size and weights and move-in paths to avoid overloading floors, elevators, etc.

- 5.3.1 **Final Connections** – Wherever possible, final connections should be designed for ease of access for construction operators. Connections required in confined spaces should be avoided wherever possible. Modular connections in confined spaces should be reviewed during safety in design, by EHS and follow Micron's confined space safety requirements.
- 5.4 Electrically, the scope demarcation between the modular constructor and the installer must be well understood and verified prior to energizing equipment. Designers can strategically plan for demarcation zones within the design to minimize risk. It is also critically important to verify the installation complies with the design.
- 5.5 Designers should clearly identify in the design documents the scope of work description interface, P&IDs, drawing, and start-up and commissioning scope of work the battery limits, interfaces, and points of connection between base build design and OSM.
 - 5.5.1 **Fall Protection** – When designing modular support structures, care should be taken to account for fall protection for the installing contractors. Fall prevention is a preferred method and should be considered during the design of large modules.

Designers can improve the safety of the installation by working to have as much scope as possible completed on the ground level and then lifted into place. Examples include pipe trestles, multi-disciplined racks, ceiling grid components, etc.

Figure 2 Risk Assessment Process from ANSI B11.0

The Risk Assessment Process Based on ANSI B11.0

6.0 General Requirements and Design Concept

"Modularization involves more than taking a traditionally designed building and cutting it up into components. Ideally, a modular-based design philosophy should be adopted from the start ensuring that various considerations including those related to geometry and material strategies, transportation logistics, and issues related to shared scope and coordination are fully integrated into the process. In general, designing with modular construction in mind from the very outset is one of the best ways to maximize the benefits and efficiencies the approach offers."-AIA Modular Design Guide.

There are three general approaches for the design of modules:

- Full A/E Design
- A/E Integration of Vendor Design
- Means and Methods of Execution

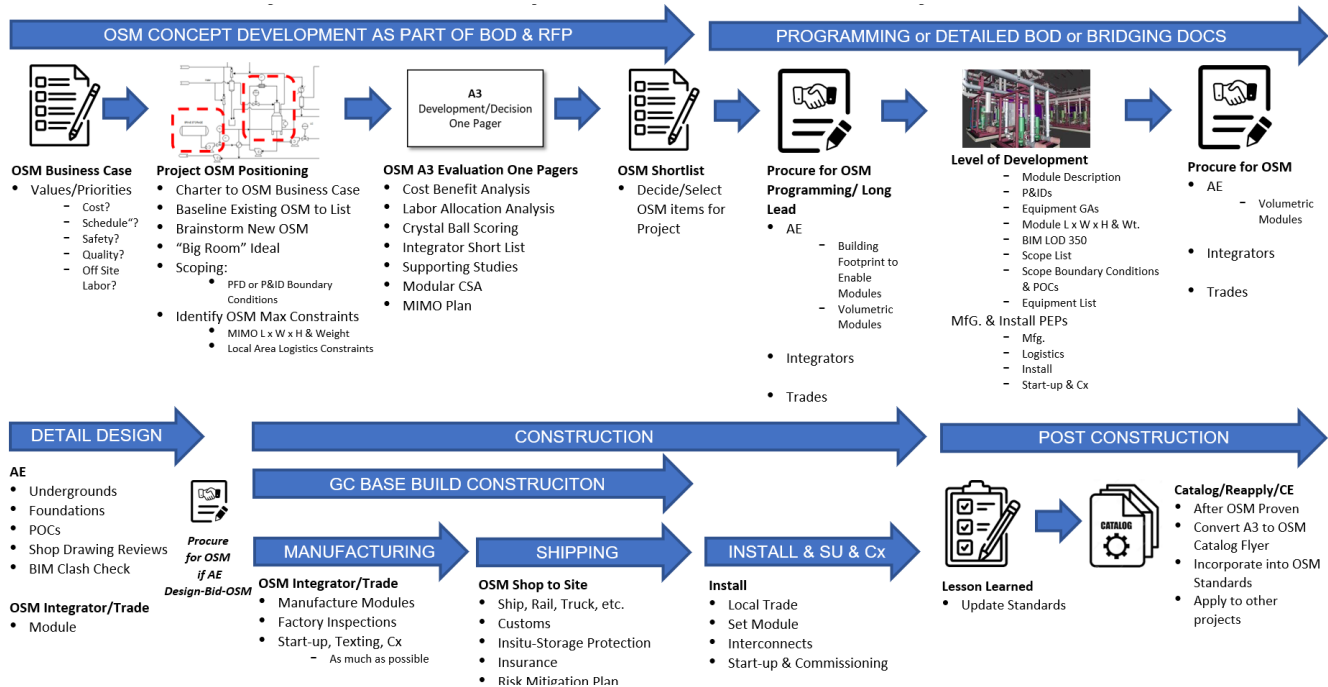
The role of the A/E is dependent on the contracting strategy and the needs of the builder. For all three approaches, a major difference between modular construction and traditional stick build is that it requires more decisions be made at an early stage in the process. Space planning, details and service integration are all typically completed earlier on modular projects because incorporating design changes to modules once fabrication has begun can be very expensive and impact schedule.

Figure 3, Source: AIA Materials Practice Guide to Modular Construction Figure 3 provides questions that should be asked when evaluating modular opportunities.



An example of a Model OSM Life Cycle Development and implementation approach for Micron, A/E and OSM supply chain GCs, Trades, and or Integrators is illustrated as follows: (figure 4)

Figure 4 OSM Life Cycle Development



6.1 Full A/E Design: Modules that will be competitively bid or will be built by a supplier without design capability will require a full A/E design. In addition, Micron may require a full design for intellectual property considerations. The full modular design will require a different level of design detail than a traditional design. Unless specified by Micron, the level of detail would include:

6.1.1 Complete set of specifications for materials and equipment, included in the module construction inclusive of all relevant systems disciplines:

6.1.1.1 General Requirements:

6.1.1.1.1 Submittal Requirements

6.1.1.1.1.1 Shop Drawings. Shop drawings are the drawings the suppliers will use to build the modules. The AE will include in the specifications any additional drawings that need to be submitted.

6.1.1.1.1.2 Material Samples. Some specifications require material samples to be submitted for review and approval. An example would be a powder coating paint sample. Any samples required should be clearly identified.

6.1.1.1.1.3 Testing Reports. Reports include pressure tests, voltage tests, quality texts, etc. All required reports should be specified.

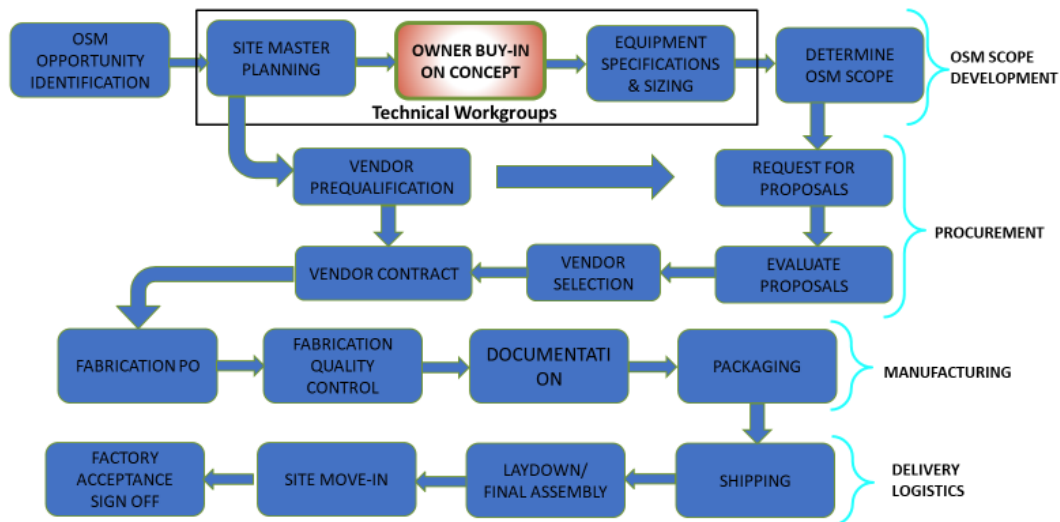
- 6.1.1.1.1.4 Module BIM to clash check and confirm fit in facility BIM. Follow the BIM Execution Plan for details.
 - 6.1.1.1.2 Alternate Means & Methods. All the modules must be compliant to local codes. Exceptions need to be submitted as alternate means and methods in a format compliant with the local Authority Having Jurisdiction.
 - 6.1.1.1.3 Start-up & Commissioning. Reports documenting startup and commissioning.
 - 6.1.1.1.4 Etc.
 - 6.1.1.2 Architectural Specifications; interior finishes, insulation, painting, exterior shells, roofs, doors etc. for volumetric, sheltered, and accessible module interiors
 - 6.1.1.3 Structural Specifications
 - 6.1.1.3.1 Seismic spec shall be included based on location
 - 6.1.1.4 Mechanical Specifications; all mechanical specifications required to construct the module
 - 6.1.1.5 Electrical Specifications; all electrical specifications required to construct the module.
 - 6.1.1.6 Process Piping Specifications; all process specifications required to construct the module
 - 6.1.1.7 Process Purity Specifications; the quality parameters required to certify the module.
 - 6.1.1.7.1 Input and output points of connections for each module will be defined
 - 6.1.1.8 Instrumentation & Controls (I&C) Specifications
 - 6.1.2 Designed to module intended location for local codes and standards, local Authority having Jurisdiction (AHJ) requirements, and local standards and testing laboratory requirements, CE, UL, etc.
 - 6.1.2.1 Designed to local specialty code requirements, pressure vessel codes, hazardous material codes, etc.
 - 6.1.2.2 Designed for local electrical requirements, Voltages, Hz, etc.
 - 6.1.2.3 Module temperature and humidity specifications for transit and in situ storage up to installation.
 - 6.1.3 PFD's and P&ID's (with flow/quality profiles) for the equipment with SOW points of connection identified and boundaries around the fabricator's scope.

- 6.1.4 Equipment general arrangement plan that includes code egress and exiting requirements, operations, service maintenance accessibility and code panel setbacks and equipment spacing.
- 6.1.5 Space planning details to clearly identify the envelope the module will fill.
- 6.1.6 Utility interconnections; piping, electrical, and control wiring.
- 6.1.7 Module Instrumentation and Controls (I&C)
- 6.1.8 A LOD450 BIM model of the module. (Note: The level of detail can vary depending on the fabricator's capabilities. It is the A/E's responsibility to ensure that the module will fit and perform as designed. A minimum of LOD300 is required regardless of the fabricator's needs to verify dimensional accuracy after fabrication is complete.
- 6.1.9 The module will be in the overall project building module and clash checked to confirm unobstructed fit of the module into the facility.
- 6.1.10 The structural supports, logistics kitting, and interconnecting utilities supports required for the module.
- 6.1.11 Connection details to show the interactions between the building and utilities.
- 6.1.12 Logistics plan to verify module load and length, width, and height dimensions are shippable to the project site.
- 6.1.13 MIMO plans highlight the route the module will take to enter the building as well as any space management considerations.
- 6.1.14 A/E to prepare construction sequence phasing diagrams and/or the prepare, with GC or Integrator pre-construction consultation, a 4D BIM construction sequence animation for more complex, larger scale modular construction projects.

Wherever possible the module designs should be reviewed and coordinated with the builder, integrators, and Micron Innovation Team.

Figure 5: Summary of the A/E Designed Modular Process described above.

FULL A/E PROCESS SUMMARY



6.2 Integrating Modules into A/E Design: Modules that will be designed and supplied by an Integrator (or other third party) should require less A/E design but more coordination support. While the A/E's deliverables are reduced, the overall system performance responsibility is still the A/E's. The deliverables include:

- 6.2.1 The A/E is to design the base building underground utilities/point of connection, foundation, equipment pad, columns, walls, pits, trenches, structural/overhead utilities/ceiling clearances and utility routings and points of connection to accommodate the module or multiple modules.
- 6.2.2 The Module Integrator provides a BIM standards compliant model for the A/E to place within the facility BIM to coordinate and confirm interfaces and for a clash free design.
- 6.2.3 Full specifications for materials and equipment included in the module construction.
- 6.2.4 P&IDs and PFDs for the equipment with SOW boundaries around the fabricator's scope.
- 6.2.5 Connection details to show the interactions between the building and utilities.
- 6.2.6 Logistics plans for staging as requested by the Integrator.
 - 6.2.6.1 Integrator is responsible for confirmation of Module logistics, shipping, in situ storage, move in paths, etc.

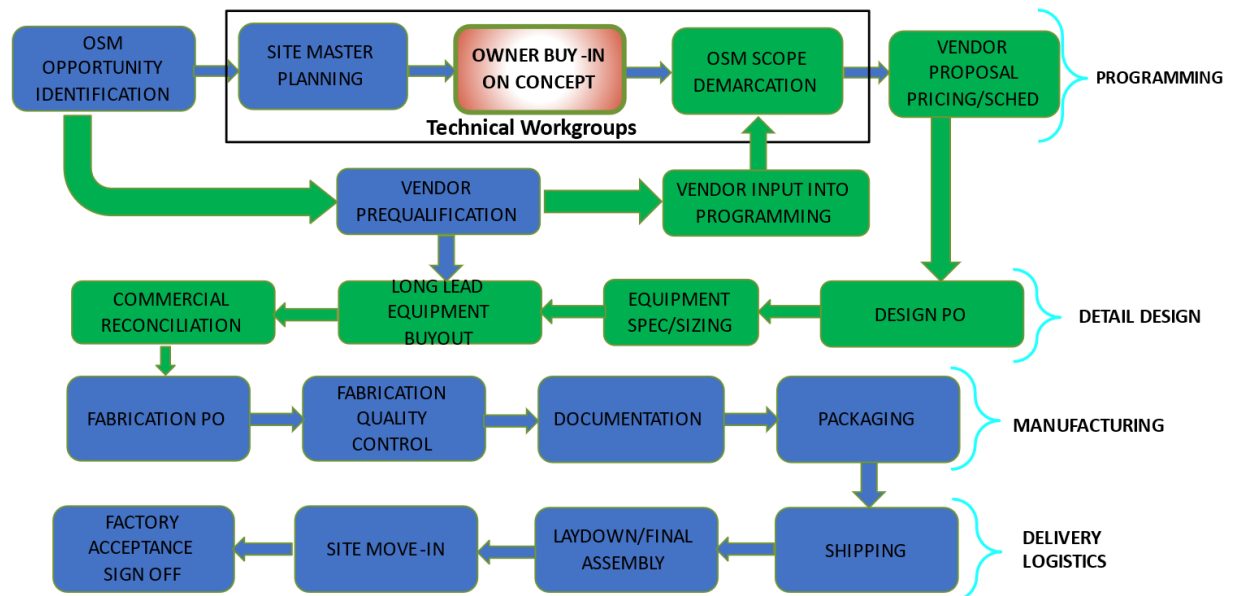
A major portion of the anticipated effort is coordinating with the Integrator to ensure they understand the nuances of the design to ensure it meets Micron's performance, reliability, safety, and quality standards.

For Integrators located outside the geography of the project, the A/E will provide code interpretation support.

6.3 Prefabricated and Preassembled Design: Prefabrication and preassembly will not require any unique design deliverables. The A/E scope for prefabrication and preassembly will be the same as general construction administration and services during construction deliverables unless otherwise specified by Micron.

Figure 6 Summarizing a Vendor designed modular process described above.

DESIGN BY INTEGRATOR PROCESS SUMMARY



7.0 Labeling Specifications

- 7.1 Labels – Equipment / Components
Per Micron Specifications
- 7.2 Labels – Piping and Tubing
Per Micron Specifications
- 7.3 Labels – Modules for Sequencing Purposes
Including QR Codes, GPS Tracking, Bluetooth tracking, etc.

8.0 Design Reviews

Micron design process includes a Basis of Design phase and a Detail Design Phase. Design reviews and OSM Integrator vendor participation will vary by Design Phase:

8.1 Basis of Design (BOD) Phase Reviews

Basis of Design is normally the first phase of design where overall scope, site master plan, general arrangements, and overall project scope is defined.

Ideally engage multiple Integration vendors for pre-qualification and bid to participate in Project Programming to program their respective module concurrent with A/E base building Positioning Phase.

- 8.1.1 **Kick-off:** Review, confirm and document in written charter OSM goals and objectives and identify major stakeholders.
- 8.1.2 **Mid-Point Review:** Halfway through BOD, A/E, GC and or Integrators to present PowerPoint process flow diagrams, equipment general arrays, systems fliers, concepts, and scopes for stakeholders review and comment.
- 8.1.3 **Final Review:** Final A/E, GC and or Integrator PowerPoint process flow diagrams, equipment general arrays, systems fliers, concepts, and scopes for stakeholders review and comment.

8.2 Detail Design Phase Reviews WS (A/E)

In the case of a full A/E module design, the A/E will prepare a typical detail design for Issue for Procurement/Manufacturing to bidding Integrators.

Ideally, the shortlist of GC's and or Integrators should participate in some of the design reviews as part of the bidding process.

- 8.2.1 **Reviewers: Review** participants should include Micron Stakeholders, GC Pre-construction representative(s) and or Integrators
- 8.2.2 **Kick-off:** Review, confirm detail design work packages and design review milestones.

8.2.3 30% Review:

- 8.2.3.1 A/E to post design review PDF drawings and Navisworks BIM 24 hours before stakeholders' review on BIM 360. The documents will include:
- 8.2.3.2 Preliminary P&ID's showing skid boundaries and connections to facility
- 8.2.3.3 Equipment/module plan view drawing
- 8.2.3.4 LOD200 BIM Model
- 8.2.3.5 A/E to host on-line and or in person 30% review of PFDs with BIM 360 review comment mark-ups and Navisworks BIM walk.
- 8.2.3.6 A/E to capture 30% design review comments in either Excel spreadsheet or database at A/E's discretion.

8.2.4 60% Review:

- 8.2.4.1 A/E to post design review PDF drawings and Navisworks BIM 24 hours before stakeholders' review on BIM 360. The documents will include:
 - 8.2.4.1.1 Completed P&ID's showing SOW boundaries with connections details and notes.
 - 8.2.4.1.2 Dimensioned plan view drawings of equipment and utilities
 - 8.2.4.1.3 LOD 300 BIM Model
 - 8.2.4.1.4 Preliminary structural plan
- 8.2.4.2 A/E to host on-line and or in person 60% review of PFDs with BIM 360 review comment mark-ups and Navisworks BIM walk.
- 8.2.4.3 All 30% design review comments are to be adjudicated by 60% review.
- 8.2.4.4 A/E to capture the new 60% design review comments.

8.2.5 90% Review:

- 8.2.5.1 A/E to post design review PDF drawings and Navisworks BIM 24 hours before stakeholders' review in BIM 360. The documents will include:
 - 8.2.5.1.1 Final P&IDs with incorporated comments from previous reviews.
 - 8.2.5.1.2 Final dimensioned plan view drawings of equipment and utilities with incorporated comments from previous reviews.

8.2.5.1.3 LOD 450 BIM Model

8.2.5.1.4 Final structural plan and construction details

8.2.5.1.5 Loose ship list with details for connecting on site as necessary

8.2.5.1.6 Specs and standards as required

8.2.5.2 A/E to host on-line and or in person 90% review of PFDs with BIM 360 review comment mark-ups and Navisworks BIM walk.

8.2.5.3 All 60% design review comments are to be adjudicated by 60% review.

8.2.5.4 A/E to capture the new 90% design review comments.

8.2.6 **100% Review & Sign-off:**

8.2.6.1 The 100% review is normally not a new review, but rather only a review of the design for adjudication of the 90% review comments and sign-off of the design by Owner stakeholders.

8.2.6.2 Once signed-off, the package is ready for A/E to IFP or IFC.

8.3 INTEGRATOR SHOP DETAIL DESIGN REVIEW

In the case where Integrator performs detail design, the Integrator design should follow the detail design review process outline above and ideally run concurrent with the A/E base building detail 30/60/90/100% review milestones.

8.4 SHOP DRAWING REVIEW

In the case where the A/E prepares a detail design of the module for bid to Integrators, the selected Integrator will prepare and submit submittals, shop drawings, material samples etc. In accordance with the general condition's submittal procedures.

9.0 Approved Manufacturers

The Pre - Qualification Questionnaire allows potential vendors to be screened based on Micron project experience, commercial viability/annual revenue, facilities available, logistical capabilities, low-cost labor GEO, unions, quality, health & safety, employee fabrication qualifications/certifications and relevant project experience. Standard prequalification procedures will be followed for OSM equipment, with the addition of the following OSM specific items:

9.1 Potential Manufacturers will be asked to provide names and location of strategic subcontractors for the equipment

9.2 Where design is required from an OSM vendor, extensive due diligence must be performed by Micron, Micron's Modular Consultant (and AE) on the proposed

vendor to validate capability of in-house engineering services and relevant experience.

- 9.3 Capabilities related to 3D scanning of the OSM equipment under fabrication will be added to the process as a part of the manufacturing progress tracking
- 9.4 The RFP from Micron to the OSM vendor will include the requirement to complete a detailed schedule of rates (SoR), developed by Micron Estimating Team, relating to the project scope for the purposes of estimating change following award. This SoR development for the OSM scope is vital and any bid not fully completing the schedule of rates will be considered invalid. This requirement must be made clear and reinforced to the bidding vendors at the RFP stage.

10.0 Manufacturing and Quality Control

This section outlines how Micron will track (any sample for tracking shall be included in appendix) the progress of vendor's manufacturing progress and quality control. This plan is issued to demonstrate systematic procedures and actions to ensure that the design, engineering, fabrication, equipment, materials, structures, and systems conform to the specific requirements and meet the intended purpose of the scope of work. This includes controlling, assuring, and documenting the technical correctness, verification of the competence of the personnel involved, and timely completion of the job. The plan is designed to assure that all scope will be performed in a highly professional manner and to the satisfaction of Micron personnel.

- 10.1 The manufacturing and quality control plan will include aspects such as:
 - 10.1.1 Monitoring progress of OSM assemblies under fabrication
 - 10.1.2 Coordination of BIM to account for OSM assemblies
 - 10.1.3 A demarcation matrix clearly defining what QC checks will happen off site versus on site, specifically around pressure testing, Helium leak checking, heat trace testing, weld inspection, insulation, and labels
 - 10.1.4 Site Inspection and Test Plan (SITP)
 - 10.1.5 Compliance plan – relating to code, specifications and standards.
 - 10.1.6 Supplier source inspections to verify function, compliance with specifications, cleanliness standards, etc.
 - 10.1.7 Tagging/labelling of components within the modules
 - 10.1.8 Handling, storage, and packaging as it relates to weather/humidity protection at the fabricator site and construction site
 - 10.1.9 Final inspection. FOAK, FAT, DV
 - 10.1.10 Schedule and Delivery
 - 10.1.11 Staging/offloading requirements (climate control)

- 10.2 To meet the above requirements, Micron will work with the OSM vendor to coordinate the following activities for manufacturing and Quality Control:
- 10.2.1 Micron will require vendors to provide the progress of the manufacturing assemblies on a weekly basis.
 - 10.2.2 BIM Standard needs to address acceptability of using OSM vendor standard BIM platform. It is un-realistic to expect a vendor to change their process to suit and A/E execution BIM platform.
 - 10.2.3 Micron will ensure that the 3D content representing all fabricated and purchased equipment is in a modeling standard as described by the BIM Standard. Specifically, 3D models will be submitted in the project standard (unless as noted above, otherwise approved by Micron).
 - 10.2.4 Micron will ensure that the BIM model accounts for any changes in as-built OSM assembly are captured.
 - 10.2.5 Micron will conduct source inspections (OSM vendor and its suppliers) monthly, with a few unannounced visits to verify adherence to Micron quality standards. Specifically, these inspections will cover aspects such as:
 - 10.2.5.1 Compliance with specifications and design standards.
 - 10.2.5.2 Micron cleanliness standards
 - 10.2.5.3 Weather protection standards
 - 10.2.5.4 Shipping and Storage Standards
 - 10.2.5.5 Compliance with local codes (as applicable)
 - 10.2.6 The OSM vendors will be required to tag and label individual pieces of infrastructure (e.g., piping, wires, ducts, etc.) before they depart the vendor facility. Accordingly, Micron will require the vendors to develop a plan for tagging and labelling within the modules/submodules as part of their Standard.
 - 10.2.7 In addition to the items mentioned above, Micron will follow standard Quality Assurance guidelines as described in the generic Quality Assurance Standard as applicable to OSM assemblies.
 - 10.2.8 It is essential that First of A Kind modules (FOAK) and components are completely reviewed and approved by Micron/Design before manufacturing proceeds. It is vital that expectations on quality/workmanship are understood by the OSM vendor and the associated supply chain of sub-contractors. Schedule will generally not allow for rework where misalignment has occurred.
 - 10.2.9 Micron will require the OSM vendors to provide an overall manufacturing and delivery tracking schedule that considers vendor's supplier dates and production queues. The detailed tracking schedule will include modules and associated submodules, as feasible. The following will be requested from the OSM vendor to get a greater visibility on progress.

- 10.2.9.1 Material pre-order dates for manufacturing/fabrication
- 10.2.9.2 Module engineering/design start and end dates
- 10.2.9.3 Module production queue and fabrication end dates.
- 10.2.9.4 Module-wise breakdown of site delivery date, including staging and storage dates
- 10.2.9.5 OSM equipment expediting method and plans
- 10.2.10 Micron will manage the OSM schedules in line with the Schedule, Micron will link vendor/supplier payments to progress of the works i.e., Earned Value. The dollar value and criteria for payments will be defined and agreed prior to contract signing
- 10.2.11 The OSM Vendor contract T&Cs will contain requirements to identify and execute required Preventative Maintenance activities on all equipment whilst it is under their possession/care.
- 10.2.12 Earned Value definition: Planned value with the budgeted amount shown through reporting periods with actual costs accrued to date. Progress payments will be made by Micron based on agreed KPI/milestones for manufacturing that are included in the OSM contract T&Cs at the bid RFP stage. The earned value will be the total budget per identified system multiplied by the % of completion agreed by the on-site PM
- 10.2.13 Micron will require the OSM vendors to produce a project organization that details responsible individuals in functions related to engineering, procurement, manufacturing, quality, start-up, EHS, and overall project management.
- 10.2.14 The OSM vendors will be required to provide a more detailed org chart.
- 10.2.15 The OSM modules will only be dispatched from the OSM vendor facility on full completion of the module with zero SORs/Punchlist items carried to the site.
- 10.2.16 Micron will propose a routine meeting and reporting structure, both face-to-face and via Teams to maintain constant communication between Micron, Designers, and the vendors. The meeting structure and agenda will be tailored to the complexity and the interdependencies of the OSM modules/projects. Micron will record meeting minutes and action required items on the Share Point.

11.0 Delivery and Logistics

This section outlines plans for shipping and logistics of the OSM assemblies. Standard logistics guidelines will be followed as a part of the OSM project.

In addition to the standard logistics guidelines, Micron will require the vendor to provide certain items related to logistics in advance of contract signing as part of vendor readiness. This will allow for proper planning of both manpower and infrastructure at the laydown area and on-site. These items are generic and will be tailored to the OSM assemblies/projects based on size, scale, and complexity.

Particular attention must be paid, and documentation developed to support the receiving and dispatching of OSM equipment at the various stages of the construction. This must include long lead equipment receiving, hand over to third party logistics/shipping and site installation. Planning for extensive observation/surveillance/monitoring and inspection of the equipment/works must be defined and understood by all parties. This critical component must be fully documented in the project OSM contract documents prior to bid.

"Delivery timetables should be carefully coordinated and adjusted according to the progress of the assembly process...

Because transportation regulations vary according to location and are handled state-by-state (in the US), if modules are manufactured in a different state than the project site, there will be a series of differing codes to comply with. In addition, the particular constraints of the project site, including the amount of space available for staging, will also affect the rate at which modules are shipped." -AIA Modular Design Guide.

In addition to above, the protection of the works relating to during assembly/fabrication, transportation, site installation and weathering must similarly have roles and responsibilities defined and path for documentation of same established in the OSM contract T&Cs.

Micron will ask the vendors to develop the following as a part of the logistics plan:

- 11.1 OSM fabrication source location
- 11.2 OSM delivery method
- 11.3 OSM transportation route planning/surveying and permitting/escorting.
- 11.4 OSM approach category
- 11.5 Skidded Framed Assembly
- 11.6 Modular
- 11.7 Other
 - 11.7.1 Container physical characteristics
- 11.8 Length
- 11.9 Width
- 11.10 Height

- 11.11 Weight
 - 11.11.1 Number of units at peak at laydown and on-site
 - 11.11.2 Assembly Scale category
- 11.12 Fully built, containerized
- 11.13 Requires yard construction
 - 11.13.1 Laydown area requirements
 - 11.13.2 Module assembly sequence on site
 - 11.13.3 Temporary laydown/site utility requirements
- 11.14 Power
- 11.15 Water
- 11.16 Gas
 - 11.16.1 Handling requirements at assembly site
- 11.17 Rigging
- 11.18 Crane
- 11.19 Forklifts
- 11.20 Other
 - 11.20.1 Packaging Requirements
- 11.21 Access To Modules
- 11.22 Crating/Weathering general protection
- 11.23 Temporary Bracing and roofing
- 11.24 Identification of the modules and tracking/tracing
- 11.25 Electronic monitoring of modules handling from dispatch to receipt.
 - 11.25.1 Environmental requirements
- 11.26 Vibration sensitivity
- 11.27 Tilt sensitivity
- 11.28 Climate control to avoid issues with condensation all through to startup on site
 - 11.28.1 Headcount requirements at assembly site
 - 11.28.2 Onsite equipment arrival controls
 - 11.28.3 Receipt verification. Inspection/Protection/Responsibilities
 - 11.28.4 Packaging/Weathering
 - 11.28.5 Identification

Micron will require the OSM vendors to develop plans for both expediting the assemblies to site as well as contingency plans for unforeseen circumstances at laydown area/site. Expediting plans will kick-in if there are delays in manufacturing or planned logistics method. Expediting will include items related to changes to delivery schedules, delivery methods (e.g., faster air/sea freights, etc.), and others as specified by vendors. Contingency plans include those to correct issues with shipment and delivery (e.g., parts not showing up, delivery of wrong parts,

breakages, quick assembly repairs, etc.) The vendors will be required to develop a strategy local to the site in Ireland to resolve these issues prior to contract signing.

The scope demarcation matrix determines how the overall logistics of the assemblies are handled and where the delineation is between OSM vendor responsibilities and responsibilities of the shipping contractor. Based on the plans and requirements from the individual vendors, Micron will coordinate the infrastructure and space requirements at the laydown area. Micron will integrate the work performed by the various stakeholders in the process until the final assemblies are delivered and signed-off by Micron.

12.0 General Contractor Roles & Responsibilities

- 12.1 The General Contractor's (GC) role for each module will vary depending on the procurement and execution strategy. For each modular opportunity, Micron's modular consultant will lead an exercise to define the scope separation. The scope separation is documented in the Modular Tracker referenced in Section 2 of this document. The GC will actively participate in this process and play an active role in coordinating work between different entities.
- 12.2 The GC will maintain the overall construction schedule for the project. Modular construction can significantly change the installation sequence as compared to traditional stick-build methods. The GC will take this into account when building the master schedule.
- 12.3 The GC will own the overall sequencing of modular construction site installations and use a collaborative planning process such as Kanban, or the Last Planner System to coordinate site readiness with the modular manufacturers. Constraints to modular manufacturing or construction should be escalated to Micron and Micron's Modular Consultant weekly. Each constraint will have an owner, a need date, a date identified, and a proposed solution.
- 12.4 The GC will be responsible for providing safety training to all Modular Integrators and owns the overall safety of the construction site. The GC will review risk assessments, pre-task plans, site incident prevention plans and other safety planning documents for modular installations as per their contract documents and Micron's safety requirements.
- 12.5 The GC owns the overall site logistics plan and will coordinate and provide staging and laydown as required for modular integrators.
- 12.6 The GC will act as a champion for modular construction. The GC will participate in the Modular Construction Work Group. Where space allows, the GC will identify areas for trade partners to pre-fabricate outside of the construction workspace.
- 12.7 The GC will coordinate crane time or other rigging as required for modular integrator deliveries. The GC will also coordinate site access for modular integrator deliveries.
- 12.8 The GC will complete QA, FAT, and Commissioning processes as required and specified in the Modular Tracker.

13.0 Testing, Commissioning, and Acceptance Testing

The Factory Acceptance Test (FAT) is a process that evaluates the equipment after the assembly process by verifying that it is built and operating in accordance with design specifications. As the name suggests, this testing is performed at the factory. The FAT is a customized testing procedure for different types of systems based on owner specifications and the tests are executed before the final installation at the facility.

- 13.1 OSM vendors will be responsible for all system commissioning aspects as per the project commissioning Standard. This includes aspects such as (but not limited to) System Integration (PLC programming for system start-up), test and balance (equipment start up), etc. system owners will assist OSM vendors in commissioning, FATs at equipment vendors where applicable, FATs/Inspections/DVs at vendor manufacturing site, and OATs (Owners Acceptance Tests)
- 13.2 FAT's will be performed per Micron Specifications and documentation provided before products ship to site for installation. Micron will deploy a project management portal that will allow all indicators, signoffs and, approvals to be managed electronically. System status will be available 24/7 in "real time".
- 13.3 FAT is conducted in order to assess discrepancies and non-conformities, as well as develop a process for how they are to be handled. Deviations or abnormalities observed during testing are documented in a report and corrected prior to shipment. (Figure 6)
- 13.4 OSM vendors will be responsible for maintenance and warranty of all infrastructure until Final Commissioning on site.
- 13.5 At a minimum, the following system control functions shall be tested and successfully proven during the System Start-up Testing & Commissioning /Acceptance Tests where applicable.
 - Motor rotation
 - Functionality of mechanical equipment
 - Leak checks (air and liquid)
 - Electrical connections
 - I&C control interface
 - Alarms and programming

Figure 7: Factory Acceptance Checklist Template

Module Assembly Factory Acceptance Test Checklist Template

Module:	
Reviewer:	
Test Date:	

General:	Scope - the tests to be performed by the manufacturer to establish acceptability of the equipment and/or skid and their conformance with the applicable drawings and specifications	
	For testers, list all applicable standards and specifications prior to the inspection	
	Codes/Standards	
	Reference Documents	
	Testing Equipment	
	Drawings Approval	
	Date:	
	FAT Checklist	
	Approval Date:	

System	Component	Reference Document	Check	Verified Y/N	Notes
		P&ID	Walk down, verify all lines and components are present and installed as per drawing. Check that the components are new, anchored sufficiently to the frame, and are built in accordance with the specifications outlined above.		
		GA	Walk down, verify all lines and components are present and installed as per drawing. Check that the components are new, anchored sufficiently to the frame, and are built in accordance with the specifications outlined above.		
		GA	Verify all components are within module size limits		
		Vessel Nameplate	Verify pressure, temperature, and material information on the nameplate is consistent with specifications/P&ID and confirm presence of ASME stamp if necessary		
		Motor Data	Verify motor manufacturer, model, and serial number. Also confirm motor classification meets specified class and division standards		
		P&ID	Verify lines hold pressure when fully assembled, record pressure at start and stop. Mark on P&ID the test boundaries and note any equipment that was tested through. Verify test is compliant with Micron Piping Spec.		
		P&ID	Verify that agitators, pumps etc rotate as expected (correct rotation, smoothly with no vibration)		
		Submittal	Verify that all system documentation is complete		
		Submittal	Failed tests or deviations should be noted		
		P&ID	Document any tests, missing components, modifications or adjustments that need to be made prior to delivery. Mark them in the notes and on the P&ID.		
		P&ID	Verify panel components, permissives, auto mode controls, manual mode controls, analog displays, sequences, reactor light operations.		
		P&ID	List all items (specifically instruments which will be removed for the ease of transportation		
		Electrical GA	If earthing/grounding is required, verify the provision for earthing/grounding connections is provided		

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14.0 Project KPIs

Micron will monitor the progress of the OSM project via specific Key Performance Indicators (KPIs). KPI's will be developed for each OSM scope (modules, skids, racks, and trestles) and will include the following items as applicable.

14.1 Vendor/Supplier Readiness

14.2 First of a kind (FOAK)

14.3 Fabrication progress

14.4 Readiness to Ship

14.5 Installation

14.6 Commissioning

14.7 Building Op Ready

The indicators will be developed on Project portal per vendor/supplier and updated on a weekly basis.

These KPIs will provide both Micron a true snapshot of OSM project progress and adherence to agreed project deliverables. These KPIs will be monitored by the project team.

15.0 Compliance to Regulations and Standards

The responsible party for design of module, either A/E, Integrator or Trade will provide Architect and or Engineer of Record design certifications, stamping, signing, per the local requirements where module is to be installed.

Module A/E of Record and or manufacturer of modules will design and manufacture the module to the local codes, standards, and owner's requirements as follow, but not limited to:

15.1 GENERAL

15.1.1 Codes and Standard as required by local AHJ.

15.1.2 ICC/MBI Standard 1200: Standard for Off-Site Construction: Planning, Design, Fabrication, and Assembly.

15.1.2.1 The International Code Council is partnering with the Modular Building Institute (MBI) in the development of a comprehensive standard to address all facets of the off-site construction process including planning; designing; fabricating; transporting; and assembling commercial and residential building elements. This includes componentized, panelised, and modularized elements.

15.1.3 ICC/MBI Standard 1210: Standard for Mechanical, Electrical, Plumbing Systems, Energy Efficiency and Water Conservation in Off-site Construction

15.1.3.1 ICC 1210-202X will address requirements for the energy efficiency and water conservation of off-site construction projects and the planning, designing, fabricating, transporting, and assembling, of commercial and residential building MEP system elements. This includes the componentization and modularization of elements of MEP systems, the incorporation of MEP systems in componentized, panelised, or modularized building elements, and the achievement of energy efficiency and water conservation requirements in off-site construction.

15.1.4 ICC G5-2019 Guideline for the Safe Use of ISO Intermodal Shipping Containers Repurposed as Buildings and Building Components

15.1.4.1 More than 30 million International Organization for Standardization (ISO) intermodal shipping containers are in use around the world today. These containers were built to ISO standards and maintained to standards defined by the International Maritime Organization's (IMO) "Convention for Safe Containers."

15.1.4.2 New or used, containers are now repurposed at a pace that makes their reuse a multi-billion-dollar global industry. Containers are regularly repurposed and converted into International Residential Code and International Building Code occupancy uses. As a building material, the applications are widely diverse as is the extent to which the container is used as a structural building element.

15.1.4.3 Local jurisdictions and state administrative programs are reacting to the growing trend of shipping container repurposing but can be behind in terms of regulations and compliance. This International Code Council Guideline is intended to help state and local jurisdictions as well as owners, architects, builders, and engineers in their assessment as to how to design, review and approve shipping containers as a building element.

15.1.5 ANSI B11.0 Safety of Machinery

15.1.5.1 This type-A standard applies to new, existing, modified or rebuilt power-driven machines, not portable by hand while working, that are used to process materials by cutting; forming; pressure; electrical, thermal or optical techniques; lamination; or a combination of these processes. This includes associated equipment used to transfer material or tooling, including fixtures, to assemble/disassemble, and to inspect or test. The associated equipment, including logic controller(s) and associated software or logic together with the machine actuators and sensors, are considered a part of the industrial machinery.

15.2 PPVC (SINGAPORE VOLUMETRIC CONSTRUCTION)

- 15.2.1 Requirements by respective agencies – The design, construction, and installation of the proposed PPVC system for building construction shall comply with the requirements by the respective agencies such as BCA, Land Transport Authority (LTA), Ministry of Manpower (MOM), National Environment Agency (NEA), PUB, The National Water Agency, Singapore Civil Defense Force (SCDF), Urban Redevelopment Authority (URA), Housing and Development Board (HDB) and JTC Corporation.
- 15.2.2 PPVC Acceptance Framework – To ensure that the different PPVC systems being used are reliable and durable, BCA has set up a PPVC Acceptance Framework consisting of building regulatory agencies as well as industry experts to evaluate the design and materials used. PPVC suppliers and manufacturers who intend to supply their PPVC systems to be used at the mandated development sites should apply for acceptance.
- 15.2.3 Sites with PPVC requirement – The use of PPVC is required for selected residential non-landed Government Land Sale (GLS) sites from 1 November 2014 onwards. Details of these sites and the mandatory PPVC requirements may be found in the Building Control (Buildability and Productivity) Regulations and in the [Code of Practice on Buildability 2017](#). More information is also available in the [circular](#) issued on 30 November 2015.

16.0 Document Control

16.1 Approval: GLOBAL_FAC_GFFT_APPR

16.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

16.3 Final View Access

All Micron Team Members at all Sites

16.4 For Retention: Use the following

Adherence to the requirements specified in

[Global Facilities - Document Control Procedures](#)

16.5 Review

Biennially (two years)

16.6 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	08/03/22

17.0 APPENDIX

17.1 Modular Factory Acceptance Test Checklist

Module Assembly Factory Acceptance Test Checklist Template

Module:	
Reviewer:	
Test Date:	

General:	Scope - the tests to be performed by the manufacturer to establish acceptability of the equipment and/or skid and their conformance with the applicable drawings and specifications		
	For testers, list all applicable standards and specifications prior to the inspection		
	Codes/Standards		
	Reference Documents		
	Testing Equipment		
	Drawings Approval Date:		
	FAT Checklist Approval Date:		

System	Component	Reference Document	Check	Verified Y/N	Notes
		P&ID	Walk down, verify all lines and components are present and installed as per drawing. Check that the components are new, anchored sufficiently to the frame, and are built in accordance with the specifications outlined above.		
		GA	Walk down, verify all lines and components are present and installed as per drawing. Check that the components are new, anchored sufficiently to the frame, and are built in accordance with the specifications outlined above.		
		GA	Verify all components are within module size limits		
		Vessel Nameplate	Verify pressure, temperature, and material information on the nameplate is consistent with specifications/P&ID and confirm presence of ASME stamp if necessary		
		Motor Data	Verify motor manufacturer, model, and serial number. Also confirm motor classification meets specified class and division standards		
		P&ID	Verify lines hold pressure when fully assembled, record pressure at start and stop. Mark on P&ID the test boundaries and note any equipment that was tested through. Verify test is compliant with Micron Piping Spec.		
		P&ID	Verify that agitators, pumps etc rotate as expected (correct rotation, smoothly with no vibration)		
		Submittal	Verify that all system documentation is complete		
		Submittal	Failed tests or deviations should be noted		
		P&ID	Document any tests, missing components, modifications or adjustments that need to be made prior to delivery. Mark them in the notes and on the P&ID.		
		P&ID	Verify panel components, permissives, auto mode controls, manual mode controls, analog displays, sequences, reactor light operations.		
		P&ID	List all items (specifically instruments which will be removed for the ease of transportation		
		Electrical GA	If earthing/grounding is required, verify the provision for earthing/grounding connections is provided		

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Global Facilities - Design & Construction Standard - General - OSM (Off-Site Manufacturing/Modularization)

Last Modified: 08/03/22

Version: 1

ID: A3YRXSD74VDV-57553043-522

17.2 Modular Tracker

Reference Number	OSM Opportunity Description	Design Package	IFC Date	Design Owner	Procurement Owner	CM Owner	Field Need Date	Main Discipline	System	FOAK (Y/N)	Expected Benefit	Execution Strategy	Supplier Pre-qual	RFP Date	PO Number	PO Date	Submittal Date	Delivery Date	Factory Test Owner	Shipping Owner	FAT Owner	Installation Owner	Commissioning Owner
1																							
2																							
3																							
4																							
5																							